Armandt Erasmus

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Education

University of Pretoria Graduation: December 2024
Bachelor of Science in Physics - Second Major: Mathematics *Pretoria, South Africa*
Bachelor of Science Honours in Physics specialising in Financial Engineering *Graduation: December 2026*

Institute and Faculty of Actuaries
Actuarial Exams: CB1, CB2, CB3, CS1, CM1, CP3 - (Expected 2027: CS2, CM2, CP2)

Experience

Actuarial Analyst/Developer Feb 2026 – Present
Miway Insurance Limited – Operational Efficiencies, Research & Development *Midrand, South Africa*

- Using Next.JS and Django, led development of a proprietary claims estimation platform incorporating GLMs, Normalised Cost of Repair (NCR), salvage and claims development methodologies to support pricing, cost estimation and reserving, deployed on AWS and delivering R24M+ in annual cost savings.
- Developed an integrated actuarial pricing and AI-powered risk assessment platform, fitting spline-based severity models and frequency GLMs to optimise risk-based premiums. Engineered a full-stack application using Next.js and Django, integrating QR-based remote self-assessments, YOLO-based computer vision damage detection and automated premium adjustments based on assessed risk.
- Engineered a 24-module AI-powered forensic fraud detection engine, integrating automated document and image manipulation detection, computer vision and probabilistic evidence scoring to quantify fraud risk and generate investigator-ready assessments within seconds. Led enterprise deployment across 3 business units and 75+ users, coordinating 14+ Manco stakeholders and delivering an estimated R7M in annual benefits.

Actuarial Analyst Jul 2025 – Feb 2026
SolveCo Actuarial Services (Pty) Ltd *Pretoria, South Africa*

- Built a centralised multi-asset financial reporting engine (VBA) covering 50+ portfolios and 10+ asset classes, standardising valuations, fair-value movements, cash-flow timing, and FX attribution to deliver IFRS, US GAAP, and GIPS-aligned TWR/MWR performance reporting.
- Developed a Python-based model validation framework that automated data preparation and testing for long-term insurance pricing models, improving governance and auditability while reducing testing timelines by 50%.

Machine Learning Engineer Jul 2023 – Jul 2024
Google Developer Student Clubs *Pretoria, South Africa*

- Led applied machine learning initiatives at the University of Pretoria, developing and delivering end-to-end ML solutions in Python, from data preprocessing and feature engineering through model training, evaluation and deployment using Scikit-learn and Streamlit for real-world business use cases.

Projects

Ratewise - Actuarial Pricing and Reserving Platform with AI Agent | *Django, Next.js, XGBoost, Claude API*

- Built a full-stack pricing and reserving platform on a standalone Python actuarial engine. It implements 10 reserving methods, including Chain Ladder, Mack and ODP bootstrap, plus pricing GLMs, GAMs and XGBoost. It runs heavy model fits asynchronously through a Celery and Redis job queue, with client polling, Parquet-backed datasets, and PostgreSQL storage for audited model runs.
- Developed an AI powered actuary agent that runs 30 actuarial tools through the same service layer as the UI. It executes on background threads with an append-only message history for prompt caching, and keeps all policy-level data away from the LLM. Deployed as five Docker Compose services with role-based access control.

Skills & Certifications

Technical: Python, Prophet, SQL (PostgreSQL, MySQL, PL/SQL), C++, R (Shiny), Excel, VBA, MATLAB, SAS

AI/ML & Data: Databricks, Scikit-Learn, TensorFlow, PyTorch, NumPy, Pandas, PySpark, Power BI, Scala

Project Management: Agile, Scrum, Stakeholder Management, Risk Management, Scheduling, Project Planning

Software Engineering: Django, React, Streamlit, Next.JS, REST APIs, Docker, Kubernetes, Git, Azure/AWS

Certifications: CFI Financial Modeling & Valuation Analyst (FMVA®), CFI Capital Markets & Securities Analyst (CMSA®), IFRS 17: Insurance Contracts - CIMA, IBM Data Science Professional Certificate, Google Project Management Certificate